

Physical Chemistry (II)

Lecture 16

2차 중간고사 안내 및 복습



2차 중간고사 안내

- 시간: 11월 13일(월) 오후 1:30~오후 2:30
- 장소: 화학관 B101호(홀수 학번), 114호(짝수 학번)
- 계산기 불필요
- 신분증 지참

시험 범위

강의 자료 기준 10강~15강

시험 문제

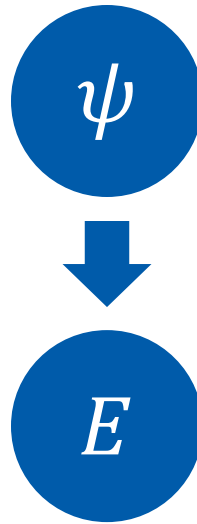
총 네 문제

- 과제 문제 혹은 그 변형(두 문제)
 - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 식 유도 및 응용(두 문제)
 - 강의 자료 기준

Overview

Hydrogen Atom

- Just H atom
- Magnetic field
- Fine structure



Beyond H Atom

- Identical particles
- H₂ molecule
- Helium atom

- Perturbation method
- Variational theory

Wavefunction of Hydrogen Atom

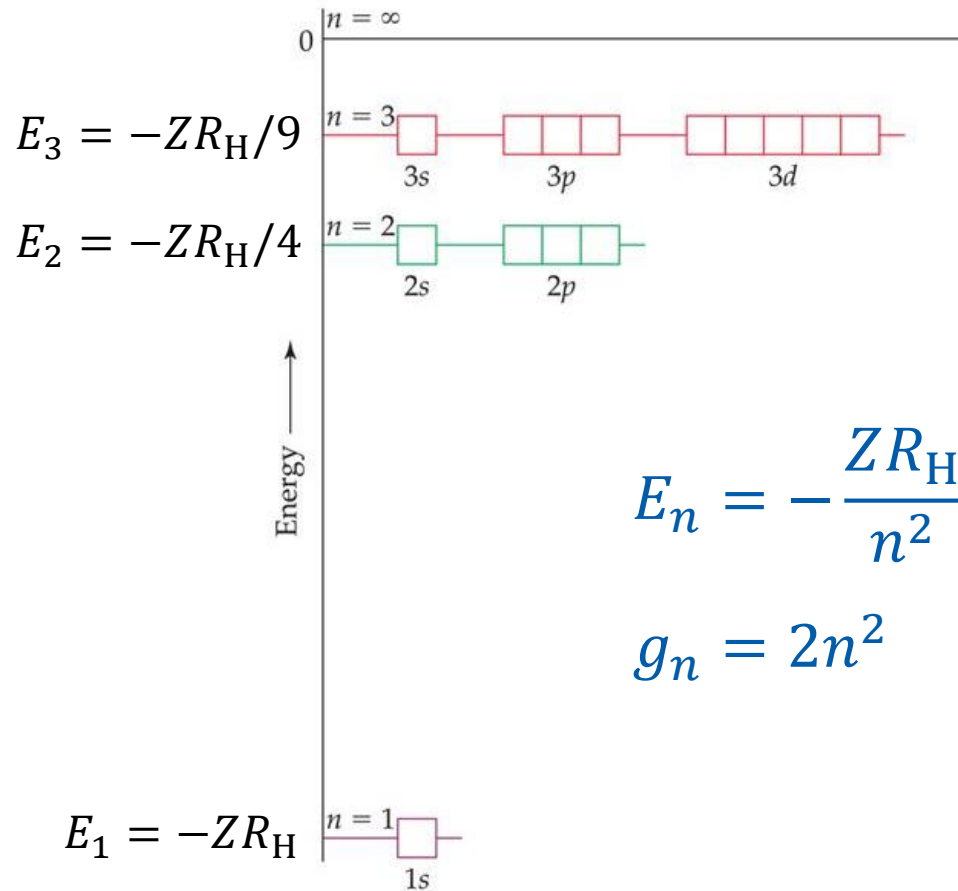
$$\frac{\psi_{nlm_l}(r, \theta, \phi)}{R_{nl}(r)Y_l^{m_l}(\theta, \phi)} |m_s\rangle$$

“spin” (PSet 4.3)

$$\hat{H}_0 = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r}$$

$$\hat{H}_0 \psi_{nlm_l}(r, \theta, \phi) |m_s\rangle = -\frac{Ze^2}{8\pi\epsilon_0 a_0} \frac{1}{n^2} \psi_{nlm_l}(r, \theta, \phi) |m_s\rangle$$

Energy Levels



Two Sets of Quantum Numbers

One can use

$$|n\ l\ m_l\ m_s\rangle = \psi_{nlm_l}(r, \theta, \phi) |m_s\rangle$$

or

$$|n\ l\ j\ m_j\rangle = \text{linear combination of } |n\ l\ m_l\ m_s\rangle$$

Different Eigenvalues

$$|n \ l \ m_l \ m_s\rangle$$

$$\hat{L}^2 |n \ l \ m_l \ m_s\rangle$$

$$= l(l+1)\hbar^2 |n \ l \ m_l \ m_s\rangle$$

$$\hat{L}_z |n \ l \ m_l \ m_s\rangle$$

$$= m_l \hbar |n \ l \ m_l \ m_s\rangle$$

$$\hat{S}_z |n \ l \ m_l \ m_s\rangle$$

$$= m_s \hbar |n \ l \ m_l \ m_s\rangle$$

$$|n \ l \ j \ m_j\rangle$$

$$\hat{L}^2 |n \ l \ j \ m_j\rangle$$

$$= l(l+1)\hbar^2 |n \ l \ j \ m_j\rangle$$

$$\hat{J}^2 |n \ l \ j \ m_j\rangle$$

$$= j(j+1)\hbar^2 |n \ l \ j \ m_j\rangle$$

$$\hat{J}_z |n \ l \ j \ m_j\rangle = m_j \hbar |n \ l \ j \ m_j\rangle$$

Hamiltonians

- Strong-field Zeeman effect $\rightarrow |n \ l \ m_l \ m_s\rangle$

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} + \frac{eB}{2\mu} (\hat{L}_z + 2\hat{S}_z)$$

- Weak-field Zeeman effect $\rightarrow |n \ l \ j \ m_j\rangle$

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r} + \frac{eB}{2\mu} \left[1 + \frac{1}{2\hat{j}^2} (\hat{j}^2 + \hat{S}^2 - \hat{L}^2) \right] \hat{j}_z$$

Fine Structure

- Good quantum numbers: $|n \ l \ j \ m_j\rangle$

$$E_{\text{fs}} = E_{\text{rel}} + E_{\text{so}} = \frac{(E_n)^2}{2mc^2} \left(3 - \frac{4n}{j + 1/2} \right)$$

- Atomic term symbols denote l , s , and j
- Selection rules: $\Delta l = \pm 1$; $\Delta j = 0, \pm 1$

Identical Particles

For a wavefunction $\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) |m_{s1}, m_{s2}, \dots, m_{sN}\rangle$,

- **Fermions:** half-integer spin (electron, proton, neutron, ...)

$$\begin{aligned} & \psi(\dots, \vec{r}_i, \dots, \vec{r}_j, \dots) | \dots, m_{si}, \dots, m_{sj}, \dots \rangle \\ &= -\psi(\dots, \vec{r}_j, \dots, \vec{r}_i, \dots) | \dots, m_{sj}, \dots, m_{si}, \dots \rangle \end{aligned}$$

- **Bosons:** integer spin (photon, helium-4 nucleus, ...)

$$\begin{aligned} & \psi(\dots, \vec{r}_i, \dots, \vec{r}_j, \dots) | \dots, m_{si}, \dots, m_{sj}, \dots \rangle \\ &= \psi(\dots, \vec{r}_j, \dots, \vec{r}_i, \dots) | \dots, m_{sj}, \dots, m_{si}, \dots \rangle \end{aligned}$$

Simple Way to Construct ψ

$$\psi = \psi_{\text{spatial}} \psi_{\text{spin}}$$

For electrons, we have two options:

- antisymmetric ψ_{spatial} $\times$ symmetric ψ_{spin}
- symmetric ψ_{spatial} $\times$ antisymmetric ψ_{spin}

Spin Wavefunctions

- Triplet: symmetric

- $|s_{\text{tot}} = 1, m_{s,\text{tot}} = 1\rangle_{\text{tot}} = |\alpha\rangle_1 |\alpha\rangle_2$

- $|1, -1\rangle_{\text{tot}} = |\beta\rangle_1 |\beta\rangle_2$

- $|1, 0\rangle_{\text{tot}} = \frac{1}{\sqrt{2}} (|\alpha\rangle_1 |\beta\rangle_2 + |\beta\rangle_1 |\alpha\rangle_2)$

- Singlet: antisymmetric

- $|0, 0\rangle_{\text{tot}} = \frac{1}{\sqrt{2}} (|\alpha\rangle_1 |\beta\rangle_2 - |\beta\rangle_1 |\alpha\rangle_2)$

Helium Atom

- Simple approximation: each electron's spatial part is represented by an one-electron orbital $\phi_i(\vec{r})$
- If the two electrons are in the same orbital (e.g. $(1s)^2$),

$$\psi_{\text{spatial}} = \phi_i(\vec{r}_1)\phi_i(\vec{r}_2): \text{symmetric}$$

$$\rightarrow \psi_{\text{spin}} = \frac{1}{\sqrt{2}}(|\alpha\rangle_1|\beta\rangle_2 - |\beta\rangle_1|\alpha\rangle_2): \text{antisymmetric}$$

Helium Atom

If the two electrons are in different orbitals (e.g. $(1s)^1(2s)^1$), we have two options:

- antisymmetric ψ_{spatial} $\times$ symmetric ψ_{spin} (triplet)

$$\psi_{\text{spatial}} = \frac{1}{\sqrt{2}} [\phi_i(\vec{r}_1)\phi_j(\vec{r}_2) - \phi_j(\vec{r}_1)\phi_i(\vec{r}_2)]$$

- symmetric ψ_{spatial} $\times$ antisymmetric ψ_{spin} (singlet)

$$\psi_{\text{spatial}} = \frac{1}{\sqrt{2}} [\phi_i(\vec{r}_1)\phi_j(\vec{r}_2) + \phi_j(\vec{r}_1)\phi_i(\vec{r}_2)]$$

Approximation Methods

We know $\hat{H}$, but cannot obtain its eigenfunctions

1. Perturbation theory

- If $\hat{H} = \hat{H}_0 + \hat{H}'$ and we know the eigenfunctions of $\hat{H}_0$

$$E_n^{(1)} = \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)} d\tau = \langle \psi_n^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle$$

2. Variational Method: ground-state energy estimation

$$E_\phi = \frac{\int \phi^* \hat{H} \phi d\tau}{\int \phi^* \phi d\tau} = \frac{\langle \phi | \hat{H} | \phi \rangle}{\langle \phi | \phi \rangle} \geq E_0$$